Linear probes reveal coding-sequence dependence in frozen DNA encoder representations

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Abstract

Motivation: Pretrained DNA encoders are often treated as general-purpose genomic representation models, but it is unclear whether their frozen embeddings linearly expose gene-level function from sequence alone. Our biological goal is to test whether raw-sequence pretraining organizes genes by functional signal; our computational goal is to measure how much of that signal can be recovered with frozen encoders and lightweight linear probes rather than DNA-model fine-tuning or large DNA-text aligners.

Results: On 3,244 human protein-coding genes from five families, frozen DNA encoders recover 5-way family signal from coding sequence (CDS). NT-v2 with concat-pooling is the strongest family classifier (macro-F1 0.828, 95% bootstrap CI [0.77, 0.87] / Cohen's κ 0.821; $\Delta\kappa$ +0.119 over a CDS 4-mer baseline of 0.672 [0.60, 0.72], CIs non-overlapping). Recoverable signal varies sharply by architecture: HyenaDNA is competitive (macro-F1 0.715) but pooling-sensitive, while GENA-LM base falls below the 4-mer baseline (0.498). For cross-modal alignment into the 1,536-d GenePT text-embedding space, DNABERT-2 recovers the most text-aligned signal (R^2 0.210 [0.19, 0.23]; ΔR^2 +0.036 over the 4-mer baseline of 0.174 [0.15, 0.19]), but alignment is uniformly weaker than discrete family classification. Replacing CDS with 196,608 bp TSS-centred windows and running all four encoders on the same regulatory substrate collapses every encoder into macro-F1 0.39–0.46 with mutually-overlapping CIs (NT-v2 0.447 [0.38, 0.51], DNABERT-2 0.455 [0.39, 0.52], HyenaDNA 0.419 [0.36, 0.48], GENA-LM 0.389 [0.33, 0.44]), and supervised Enformer trunk features (macro-F1 0.545) recover more TSS signal but remain far below CDS NT-v2 — the substrate, not the encoder, is the dominant variable. The benchmark therefore supports a narrow claim: frozen DNA encoders linearly expose coding-sequence family signal but not generic gene function from arbitrary genomic context.

Availability and implementation: https://github.com/TBD/dna-to-text, MIT license, Python $\geq$3.10, PyTorch, transformers, scikit-learn, UMAP.

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Supplementary information: Supplementary data are available in the appendix.

1 Introduction

Pretrained DNA encoders such as DNABERT-2 (Zhou *et al.*, 2023), Nucleotide Transformer (Dalla-Torre *et al.*, 2024), GENA-LM (Fishman *et al.*, 2025), and HyenaDNA (Nguyen *et al.*, 2023) learn dense representations of genomic sequence with different tokenisers, architectures, context windows, and pretraining corpora. Their frozen embeddings are often treated as biologically meaningful features, but strong downstream performance after fine-tuning does not by itself show what is already linearly available in the frozen representation. Linear probing (Alain and Bengio, 2016) is the standard tool for that narrower question.

Our biological goal is to test whether DNA encoders trained on raw sequence implicitly organize genes by functional signal. Our computational goal is to ask how much of that signal can be recovered with a lightweight frozen-encoder and linear-probe pipeline, rather than by fine-tuning the DNA model or training a large DNA-text alignment system.

The key ambiguity is therefore not simply whether a DNA foundation model is useful. It is whether a frozen DNA representation contains gene-level functional signal beyond local nucleotide composition, and whether that signal depends on the biological context supplied to the encoder. Recent multimodal systems already combine DNA and language: Omni-DNA uses autoregressive genomic pretraining followed by multi-task fine-tuning for tasks including DNA-to-function text generation, while shared-tokenizer and other DNA-language fusion methods train alignment or integration modules over DNA-text pairs (Li *et al.*, 2025, 2026). These systems show that sequence-text modeling is possible, but they require substantial additional multimodal training. GenePT represents each gene by embedding a natural-language summary of that gene, yielding a text-derived gene vector that can serve as a cross-modal target for sequence models (Chen and Zou, 2023). Our question is narrower: whether frozen DNA embeddings already align with gene-summary text embeddings under lightweight linear probes, compositional controls, shuffled-label anti-baselines, pooling sweeps, and matched coding-sequence versus regulatory-context comparisons.

We probe four frozen self-supervised DNA encoders on two tasks: 5-way gene-family classification and Ridge regression into a 1,536-d GenePT summary-embedding target (Chen and Zou, 2023). The corpus contains 3,244 human protein-coding genes from transcription-factor,

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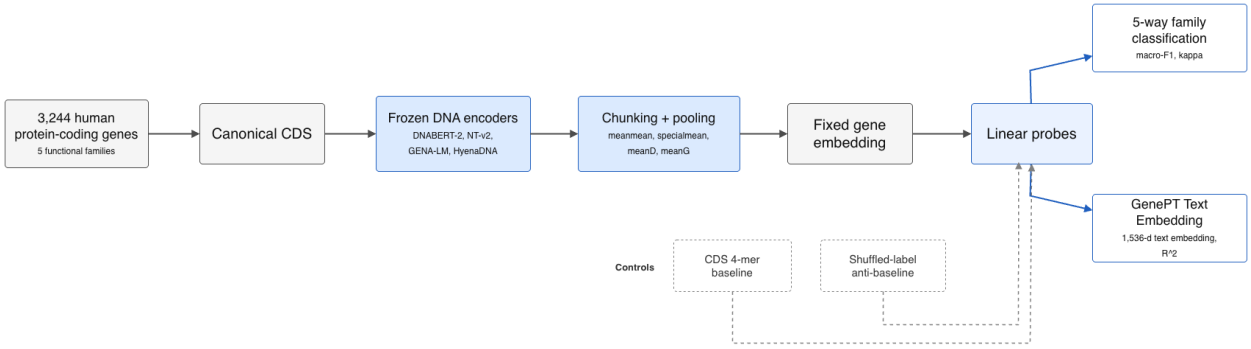

Fig. 1. Study design. Human protein-coding genes are represented by canonical CDS, encoded with frozen DNA language models, reduced to fixed gene embeddings by chunking and pooling, and evaluated with linear probes for family classification and Ridge-to-GenePT alignment. CDS 4-mer and shuffled-label controls test compositional and pipeline-leak baselines.

GPCR, kinase, ion-channel, and immune-receptor families. Each CDS comparison is anchored to a 4-mer baseline and a shuffled-label anti-baseline. We then ask whether the same family signal is recoverable from 196,608 bp windows centred on transcription start sites (TSS), using matched TSS 4-mers, all four encoders evaluated on the same TSS windows, and Enformer (Žiga Avsec *et al.*, 2021) as a supervised sequence-to-function comparator. This design separates the paper's supported claim from the tempting but broader one: the evidence is for coding-sequence family signal, not generic gene function from arbitrary genomic sequence.

2 Methods

Dataset. We curated 3,244 human protein-coding genes from five functional families: transcription factors (1,743), G-protein-coupled receptors (591), protein kinases (558), ion channels (198), and immune receptors (154). Genes were selected by regex match against the HGNC `gene_group` field (Seal *et al.*, 2023), intersected with the GenePT v2 gene-symbol set (Chen and Zou, 2023), and deduped first-family-wins in the assignment order kinase → transcription factor → ion channel → GPCR → immune receptor (so a gene matching multiple HGNC groups is counted under the earliest-priority family — e.g., a HIPK kinase that also has a TF role is counted as a kinase). The Family column in Table 1 is therefore size-ranked but not assignment-ranked. Canonical coding sequences (CDS) were retrieved from Ensembl (Harrison *et al.*, 2024). The corpus was split 70/15/15 stratified by family with seed 42 and frozen for all downstream runs (Table 1).

Table 1. Dataset composition and frozen split.

Family	Total	Train	Val	Test
Transcription factor	1,743	1,220	262	261
GPCR	591	414	88	89
Kinase	558	390	84	84
Ion channel	198	138	30	30
Immune receptor	154	108	23	23

Frozen 70/15/15 split.

Encoders. We compared four frozen self-supervised DNA encoders on canonical CDS. **DNABERT-2** (Zhou *et al.*, 2023) is a 117 M-parameter BERT-style model (`zhihan1996/DNABERT-2-117M`) with byte-pair encoding, 768-d hidden states, a 512-token context window, and `[CLS]` / `[SEP]` boundary tokens. **Nucleotide Transformer v2 multi-species 100M** (Dalla-Torre *et al.*, 2024) (`InstaDeepAI/nucleotide-transformer-v2-100m-multi-species`, hereafter NT-v2) is an ESM-style masked-language model with fixed non-overlapping 6-mer tokens, 512-d hidden states, a 2,048-token context window, and a `<cls>` marker. **GENA-LM base** (Fishman *et al.*, 2025) (`AIRI-Institute/gena-lm-bert-base-t2t`) is a 110 M-parameter BERT-12L masked-language model with 768-d hidden states, 12 attention heads, a 32,000-token BPE vocabulary, and a 512-token context window corresponding to roughly 4.5 kb after BPE compression; the `t2t` checkpoint was trained on the human T2T assembly with 1000 Genomes SNP augmentation. **HyenaDNA large** (Nguyen *et al.*, 2023) (`LongSafari/hyenadna-large-1m-seqlen-hf`) is an attention-free causal next-nucleotide model based on Hyena sequence operators, with single-nucleotide tokens, 8 layers, 256-d hidden states, approximately 6.6 M parameters, and a context window up to 1M nucleotides; HyenaDNA was pretrained on the human reference genome. Each canonical CDS was chunked and each chunk was wrapped with the model's standard special tokens (`[CLS]`/`[SEP]` for DNABERT-2 and GENA-LM, `<cls>` for NT-v2; HyenaDNA is causal and has no learned summary token), encoded in PyTorch via Hugging Face Transformers (Wolf *et al.*, 2020), and reduced to a single fixed-dimensional vector for probing. All encoder weights were frozen.

Pooling sweep. Because many CDS exceed encoder context windows, each gene must be represented by a fixed-size vector after chunking. We therefore evaluated simple parameter-free pooling rules rather than training an additional sequence aggregator. Mean, max, and CLS pooling are standard reductions for contextual encoder embeddings (Devlin *et al.*, 2019; Reimers and Gurevych, 2019); `specialmean` is the corresponding mean-pooling control that retains model boundary/special-token positions; `meanD` and `meanG` add coarse first/last/global summaries so long genes are not represented only by an unordered average. The full cached CDS matrix for 5-way family classification and Ridge-to-GenePT regression is reported in Supplementary Table S1.

Variant	Gene-level reduction
`meanmean`	average token embeddings, then average chunks
`specialmean`	average all tokens including special tokens, then average chunks
`maxmean`	max-pool token embeddings, then average chunks
`clsmean`	use CLS-position embedding, then average chunks
`meanD`	concatenate first chunk, last chunk, and chunk mean
`meanG`	`meanD` plus a max-over-chunks summary

TSS context and Enformer comparator. To test whether the same family signal is recoverable from regulatory context, we constructed 196,608 bp genomic windows centred on each gene's strand-specific transcription start site. These windows are much larger, mostly noncoding,

and regulatory rather than protein-domain-rich, so we expected weaker family-label recovery than from CDS while gaining a direct CDS-versus-regulatory-context ablation.

We ran all four self-supervised DNA encoders (NT-v2, DNABERT-2, GENA-LM, HyenaDNA) over the same 196,608 bp TSS windows with the CDS pooling rules and probe protocol, plus a matched TSS-window 4-mer baseline. For each encoder we report the best-pool result against the TSS 4-mer baseline. We also included Enformer (Žiga Avsec *et al.*, 2021), a supervised DeepMind sequence-to-function model rather than a DNA language model. Enformer consumes one-hot 196,608 bp genomic windows, applies seven convolution/pooling blocks followed by 11 transformer blocks, and predicts 5,313 human plus 1,643 mouse genomic tracks over 896 central 128-bp bins. This supervised target makes Enformer the natural TSS-to-function comparator rather than a CDS encoder. Trunk embeddings are internal Enformer tower representations before the organism-specific output heads. From the 896 $\times$ embedding-dim trunk we computed three summaries: `trunk_global` (mean across all 896 bins), `trunk_center` (mean across the central 16 bins, configurable via a `-center-bins` flag), and `tracks_center` (mean across the same central 16 bins of the 5,313 human and 1,643 mouse track predictions, i.e. the post-head supervised regulatory-track features).

Probes. For the 1,536-d GenePT text-embedding regression target we trained multi-output ridge regression with `scikit-learn` (Pedregosa *et al.*, 2011), on top of `NumPy` (Harris *et al.*, 2020) and `SciPy` (Virtanen *et al.*, 2020); ridge was chosen over reduced-rank or partial-least-squares variants for its closed-form efficiency on the high-dimensional target and its alignment with the standard linear-probing protocol. For classification we trained multinomial logistic regression. Regularisation (α for ridge, C for logistic) was swept on the validation split over a decade grid, the probe was refit on the train+val concatenation at the selected hyperparameter, and metrics were evaluated *once* on the held-out test split; selected hyperparameters are reported per cell in Supplementary Table S1. Linear probing follows the protocol of Alain and Bengio (2016). For the headline cells we also report 95% confidence intervals on macro-F1, κ, and macro R^2 obtained from a 1,000-iteration bootstrap of the held-out test split (stratified by family for classification, i.i.d. by gene for regression); the bootstrap quantifies sampling uncertainty in the test composition only and not hyperparameter or split-seed variability.

Baselines and controls. The *k-mer composition* baseline projects each CDS, or each TSS window for the context ablation, to a 256-d L_1-normalised 4-mer histogram and runs the same probe recipe; it is encoder-independent and serves as the compositional floor. The *shuffled-label anti-baseline* permutes family labels in train+val only (test labels remain real) and tests pipeline honesty: chance-corrected agreement should collapse to zero. For the regression probes, an analogous *shuffled-Y anti-baseline* permutes the 1,536-d GenePT vectors in train+val only and re-runs the full α sweep on the shuffled validation split (rather than reusing the α selected from the real run), so any non-trivial test R^2 would imply leakage in either the data split or the hyperparameter selection.

Evaluation metrics. For 5-way family classification, the primary metric is macro-F1, with Cohen's κ (Cohen, 1960), accuracy, and confusion matrices reported as supporting diagnostics for class imbalance. For Ridge-to-GenePT regression, the primary metric is macro-averaged held-out R^2 across the 1,536 GenePT dimensions. These two endpoints define the main paper claims.

3 Results

3.1 CDS encoders separate gene families

On canonical CDS, NT-v2 is the strongest encoder for 5-way family classification: `meanD` pooling reaches macro-F1 0.828 (95% bootstrap CI [0.77, 0.87]) / Cohen's κ 0.821 ([0.77, 0.86]), a $\Delta\kappa$ of +0.119 over the CDS 4-mer baseline (macro-F1 0.672 [0.60, 0.72] / κ 0.702 [0.64, 0.75]; CIs non-overlapping on both metrics; Table 2). For this imbalanced 5-way task, κ rewards majority-class agreement (TFs are 53% of the corpus) while macro-F1 weights every class equally, so an encoder that mainly tracks the TF majority shows κ above macro-F1 (as does the 4-mer baseline: κ 0.702 versus macro-F1 0.672), whereas one that also recovers minority families inverts the inequality (as does NT-v2: κ 0.821 versus macro-F1 0.828). Per-class F1 confirms minority recovery: NT-v2 `meanD` reaches 0.93 (TF), 0.94 (GPCR), 0.79 (kinase), 0.68 (ion), 0.80 (immune), versus the 4-mer baseline at 0.88, 0.91, 0.66, 0.39, 0.47 — the encoder gain is concentrated in the minority classes (immune $+0.33$, ion $+0.29$). The shuffled-label anti-baseline collapses to macro-F1 0.208 / κ 0.048, so the family signal is not split leakage or probe bookkeeping. HyenaDNA is competitive with DNABERT-2, while GENA-LM base falls below CDS 4-mer composition; thus, "DNA language model" is not a sufficient explanation for success.

Table 2. Best cached 5-way family classification cell. Best non-control value is bolded.

Source	Pool	F1	κ	$\Delta\kappa$	Acc.
Shuffled labels	—	0.208	0.048	-0.654	0.435
CDS 4-mer	—	0.672	0.702	0.000	0.817
DNABERT-2	`meanD`	0.738	0.723	+0.020	0.823
NT-v2	`meanD`	**0.828**	**0.821**	**+0.119**	**0.887**
GENA-LM base	`clsmean`	0.498	0.478	-0.225	0.686
HyenaDNA large	`meanG`	0.715	0.694	-0.008	0.809

Test split metrics; full pooling sweep in the appendix.

3.2 Cross-modal alignment is weaker than family classification

Linear projection into the 1,536-d GenePT text-embedding space recovers a smaller signal than family classification (Table 3). We report macro R^2 rather than cosine because GenePT space has a high anisotropy floor: the shuffled-Y anti-baseline still attains cosine 0.913, so cosine is a poor separator and R^2 is the honest metric. DNABERT-2 `meanG` is the best Ridge probe (R^2 0.210, 95% bootstrap CI [0.19, 0.23]), only +0.036 above the CDS 4-mer baseline (R^2 0.174 [0.15, 0.19]; CIs non-overlapping). NT-v2 reaches R^2 0.193, HyenaDNA 0.182, and GENA-LM 0.117, so the strongest CDS family separator is not necessarily the representation most aligned with text-summary geometry.

Table 3. Best cached Ridge-to-GenePT cell. Best non-control value is bolded.

Source	Pool	R^2	Δ	Cos.
Shuffled labels	—	-0.000	-0.175	0.913
CDS 4-mer	—	0.174	0.000	0.931
DNABERT-2	`meanG`	**0.210**	+0.036	**0.934**
NT-v2	`meanmean`	0.193	+0.019	0.932
GENA-LM base	`meanmean`	0.117	-0.057	0.925
HyenaDNA large	`specialmean`	0.182	+0.008	0.931

Δ is relative to the CDS 4-mer baseline; base/`meanmean` ties are shown as `meanmean`.

The modest R^2 values constrain the biological claim: GenePT summaries are heterogeneous and not limited to protein-family

membership. They are also broadly rather than narrowly carried: for DNABERT-2 `meanG` CDS, 1,527 of the 1,536 GenePT dimensions have positive test R^2, the median per-dim R^2 is 0.190 (within 0.02 of the 0.210 macro mean), and recovering 50% of the summed R^2 requires the top 429 dimensions (28% of the embedding space); the other three encoders show the same broad shape one band lower (Supplementary Figure A5). The headline cross-modal signal is therefore distributed across most of the GenePT space rather than concentrated in a handful of well-predicted summary axes. The joint view in Figure 2 makes the encoder tradeoff explicit: NT-v2 is the strongest family classifier, whereas DNABERT-2 is most Ridge-aligned.

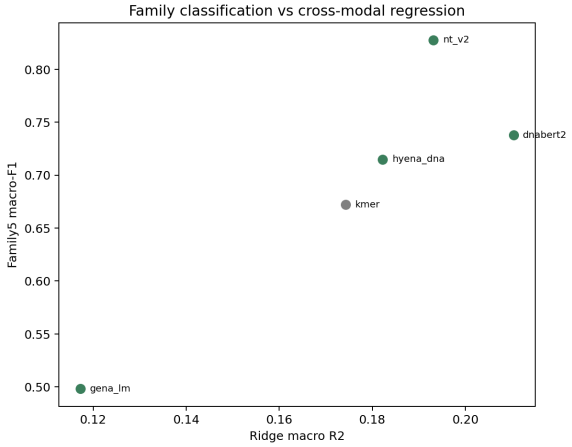

Fig. 2. 5-way family classification versus Ridge-to-GenePT alignment. NT-v2 is strongest for classification; DNABERT-2 is strongest for Ridge.

3.3 Pooling controls representation geometry

Pooling choice changes apparent representation quality (Figure 3). Concrete examples bracket the range: NT-v2 spans macro-F1 0.586 (`maxmean`) to 0.828 (`meanD`); HyenaDNA's `clsmean` collapses to macro-F1 0.140 / κ 0 because the causal model has no BERT-style trained summary token, while its concat-style `meanG` reaches 0.715. Concat-style summaries (here `meanD`, the first/last/global-mean concatenation, and `meanG`, which adds a max-over-chunks term) help both NT-v2 and DNABERT-2; `specialmean` is nearly indistinguishable from `meanmean` on CDS, indicating that special-token inclusion is a robustness check rather than a main performance driver. The full encoder $\times$ pooling matrix is in Supplementary Table S1; Tables 2 and 3 report best-cell-per-encoder selections from this 24-cell sweep. The readout therefore interacts with architecture strongly enough that the same encoder can appear weak or strong depending on pooling choice; pooling is not a neutral implementation detail.

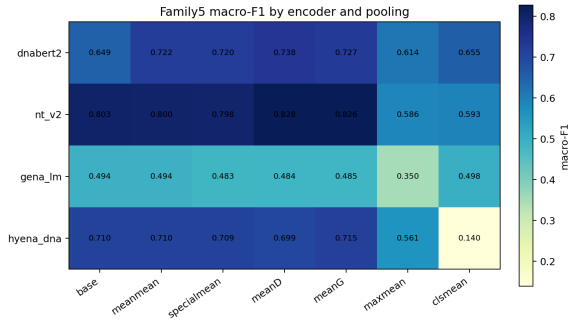

Fig. 3. 5-way family classification macro-F1 across cached encoder-pooling cells. Concat-style summaries help NT-v2 and DNABERT-2; special-token-inclusive mean pooling is largely neutral; CLS-style pooling is architecture-dependent.

3.4 Genomic context controls recoverable family signal

Holding the gene set fixed and replacing the CDS substrate with 196,608 bp TSS-centred windows collapses family signal across *every* self-supervised encoder we tested (Figure 4). NT-v2 reaches macro-F1 0.828 [0.77, 0.87] on CDS but only 0.447 [0.38, 0.51] on TSS; DNABERT-2 0.738 [0.68, 0.79] on CDS but 0.455 [0.39, 0.52] on TSS; HyenaDNA 0.723 [0.66, 0.78] on CDS but 0.419 [0.36, 0.48] on TSS; GENA-LM 0.500 [0.44, 0.56] on CDS but 0.389 [0.33, 0.44] on TSS. All four encoders cluster in macro-F1 0.39–0.46 on TSS with mutually-overlapping 95% bootstrap CIs, so the substrate effect is encoder-general rather than an NT-v2 artefact. The drop is expected: a 196 kb TSS window is a much larger, mostly noncoding regulatory substrate, so family-specific CDS signal is diluted by context not directly tied to protein family. The TSS result is nevertheless positive: every encoder beats the TSS 4-mer baseline (macro-F1 0.247 [0.23, 0.27]) with non-overlapping CIs, indicating that frozen DNA language embeddings recover nontrivial regulatory-context structure beyond local composition. Some of this TSS signal is mechanistically expected: family-typical regulatory motifs cluster near family-typical promoters (e.g., TF binding-site enrichment near TF genes), so the 0.4-ish TSS macro-F1 partly reflects family-correlated cis-regulatory structure rather than purely independent regulatory information. Supervised Enformer trunk features improve further (macro-F1 0.545; Ridge R^2 0.142), as expected for a model trained on long-window sequence-to-function targets, but remain below the CDS encoders. The benchmark therefore supports a narrower claim: frozen DNA encoders recover coding-sequence family information, with recoverability controlled by genomic substrate uniformly across architectures. The CDS advantage over 4-mers is consistent with the encoder distinguishing protein-family-relevant coding-sequence structure rather than treating the gene as an unordered bag of short motifs.

The same encoder-general pattern holds for the GenePT text-embedding regression target. On TSS, DNABERT-2 (macro-R^2 0.122 [0.10, 0.14]) and NT-v2 (0.117 [0.09, 0.14]) lead, followed by HyenaDNA (0.085 [0.07, 0.10]) and GENA-LM (0.059 [0.04, 0.07]); all four sit above the 4-mer TSS baseline (0.041 [0.03, 0.05]) with non-overlapping CIs and well below their CDS counterparts. The encoder ranking on TSS differs slightly from CDS (DNABERT-2 nudges ahead of NT-v2 once context dominates over coding sequence), but no two encoders are separated by a non-overlapping CI on TSS, so we read the ranking as essentially tied rather than as a substantive architecture effect.

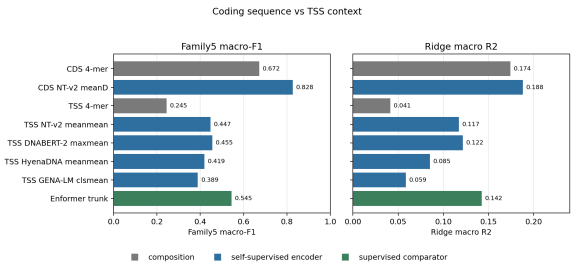

Fig. 4. Context ablation across four self-supervised DNA encoders plus the supervised Enformer comparator. CDS encoders span macro-F1 0.50–0.83; switching the substrate to 196,608 bp TSS-centred windows collapses every encoder into the 0.39–0.46 band, well above the TSS 4-mer baseline (0.247) but well below CDS NT-v2. Encoder choice on TSS is statistically immaterial (mutually-overlapping CIs); the substrate is the dominant variable.

4 Discussion

Supported claim. The central result is that frozen DNA encoders can expose coding-sequence family signal to a lightweight linear probe. This is not a trivial success against chance: the CDS 4-mer baseline is already strong, and the shuffled-label anti-baseline collapses as expected. NT-v2 nevertheless improves over CDS 4-mers by $\Delta\kappa$ +0.119 on 5-way family classification, showing that at least one frozen encoder contains family-discriminative structure beyond unordered short-motif composition. The evidence therefore supports a narrow claim: linearly recoverable family signal in canonical CDS, not general gene-function prediction.

Genomic context. The CDS-versus-TSS ablation is the cleanest guardrail on interpretation because it holds the gene set fixed while changing the input substrate. Running all four self-supervised encoders on the same 196,608 bp TSS windows tightens the claim: every encoder drops by 0.27–0.38 macro-F1 versus its CDS result, and all four cluster in macro-F1 0.39–0.46 on TSS with mutually-overlapping 95% bootstrap CIs. That collapse is biologically coherent: these labels are mostly protein-family labels, while TSS windows contain promoter, regulatory, untranslated, intronic, and broader genomic context. Every encoder still beats the TSS 4-mer baseline (0.247) with non-overlapping CIs, so the regulatory windows are not empty; they expose indirect correlates of family membership but do not match CDS encoders on protein-family labels. The substrate, not the encoder, is the dominant variable: the substrate-dominance claim is now multi-encoder and not an NT-v2 artefact. Enformer trunk features remain the strongest TSS comparator at macro-F1 0.545, consistent with its supervised long-window training objective.

Model and pooling specificity. The encoder comparison argues against treating DNA language models as interchangeable feature extractors. NT-v2 is the clearest classification winner, DNABERT-2 is strongest for Ridge-to-GenePT regression, HyenaDNA is competitive on family classification, and GENA-LM base falls below CDS 4-mer composition on both probes. GENA-LM is in this sense an informative negative: a pretrained DNA language model can underperform a 256-d 4-mer composition baseline on the same corpus and probe recipe, so model-name optimism is not a substitute for benchmarked representation analysis. Plausible mechanisms for GENA-LM's underperformance — none verified here — include the BPE tokeniser's loss of codon-frame information after segmentation, the relatively short $\sim$4.5 kb effective context after BPE compression cutting many CDS into a single chunk, and a pretraining objective on the human T2T assembly that may not directly reward CDS-level structure. Pooling is likewise part of the representation interface rather than a neutral detail: concat-style summaries (`meanD`, `meanG`) help NT-v2 and DNABERT-2, CLS-style pooling collapses for HyenaDNA — a causal model with no trained summary token — and `maxmean` hurts every encoder we tested. Special-token treatment is the milder cousin of pooling here: `specialmean` is statistically indistinguishable from `meanmean` on CDS for every encoder, so for the architectures and chunk sizes studied, pool-time inclusion of `[CLS]`/`[SEP]`/`<cls>` positions matters less than concat-versus-flat aggregation. The unifying point is that protocol details (pooling rule, special-token handling) interact with architecture, so a model can appear weak because the probe asks the wrong summary vector to carry gene-level information.

Cross-modal signal. Ridge regression into GenePT space gives a complementary but weaker view of the same representations. The best cross-modal gain is modest (ΔR^2 about +0.04 over CDS 4-mers), and the model ranking differs from classification: DNABERT-2 aligns best with GenePT summary geometry while NT-v2 best separates families. This split is not a contradiction. Family classification asks whether embeddings expose a discrete protein-family label; GenePT regression asks whether sequence features linearly predict a high-dimensional text-summary embedding. In this benchmark, cross-modal alignment is detectable but partial.

5 Conclusion and Study Limitations

5.1 Conclusion

Frozen DNA encoders can recover coding-sequence family signal beyond simple composition, but the strength and meaning of that signal depend on genomic context, model recipe, pooling, and the target used to define "function." The benchmark therefore supports a precise conclusion: frozen DNA representations can be useful linearly readable CDS feature extractors, but these results should not be read as generic recovery of gene function from arbitrary genomic sequence.

5.2 Study Limitations and Next Steps

Several limitations follow directly from the design. The corpus is human-only, spans $\sim$3,200 genes, and uses five broad protein-family categories that all carry strong, well-characterised domain motifs (kinase domains, GPCR transmembrane patterns, ion-channel selectivity filters, TF DNA-binding domains, immune-receptor domains); a more diverse ontology or families with weaker domain signatures might yield smaller encoder-versus-k-mer gaps. Family membership was deduplicated first-family-wins across overlapping HGNC groups, but the train/test split was random-stratified at the gene level rather than clustered by sequence similarity, so paralogous genes in the same family can sit on both sides of the split; the headline numbers should be interpreted under this potential leakage. The TSS context arm now covers all four self-supervised encoders plus the supervised Enformer comparator, but it is still restricted to a single 196,608 bp window centred on the strand-specific TSS, and Enformer's pretraining tracks include many of the gene loci it is evaluated on, so the substrate-dominance claim is constrained to this window choice and to Enformer-overlapping loci. HyenaDNA is also substantially smaller than the BERT-style encoders (256-d hidden, $\sim$6.6 M parameters), so its competitive but lower headline result confounds architecture, capacity, and pretraining objective.

On the analysis side, headline cells now carry 1,000-iteration bootstrap CIs of the held-out test split (Methods), but these capture test-composition sampling uncertainty only and not hyperparameter or split-seed variability; paired permutation tests and seed-variance estimates are still absent, and the minority test classes (immune n=23,

ion n=30) make per-class macro-F1 components noisy. Best-cell-per-encoder reporting from the 24-cell pooling sweep introduces a multiplicity that is not corrected; the full grid is reported in Supplementary Table S1. More controlled conclusions would require matched encoders that vary one axis at a time, paralog-aware splitting (e.g., MMseqs2 or CD-HIT clustering at a stated identity threshold), additional organisms, noncoding and regulatory targets, promoter/enhancer-specific labels, per-layer probes, paired permutation tests, and a substrate sweep over multiple TSS window sizes rather than the single 196,608 bp window used here.

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A Supplementary Tables and Figures

The appendix reports curated cached artifacts used to support the main 5-way family classification and Ridge claims. All tables use the frozen 70/15/15 split and the same validation-tuned probe protocol as the main text.

A.1 5-Way Family Classification and Ridge Headline Tables

Table A1. Best cached 5-way family classification cell per encoder plus baselines.

Encoder	Pooling	Macro-F1	κ	$\Delta\kappa$	Accuracy
CDS 4-mer	baseline	0.6722	0.7024	0.0000	0.8172
DNABERT-2	meanD	0.7380	0.7226	+0.0202	0.8234
NT-v2	meanD	0.8275	0.8214	+0.1190	0.8871
GENA-LM base	clsmean	0.4982	0.4776	-0.2248	0.6858
HyenaDNA large	meanG	0.7149	0.6944	-0.0080	0.8090
Shuffled labels	anti-baseline	0.2078	0.0482	-0.6542	0.4353

Source: `analysis/tables/main_family5.csv`.

Table A2. Best cached Ridge-to-GenePT cell per encoder plus CDS 4-mer baseline.

Encoder	Pooling	R^2 macro	Δ vs 4-mer	Mean cosine
Shuffled labels	–	-0.0004	-0.1746	0.9130
CDS 4-mer	baseline	0.1743	0.0000	0.9306
DNABERT-2	meanG	0.2104	+0.0361	0.9340
NT-v2	meanmean	0.1932	+0.0189	0.9324
GENA-LM base	base	0.1173	-0.0569	0.9251
HyenaDNA large	specialmean	0.1822	+0.0080	0.9313

Source: `analysis/tables/main_regression.csv`.

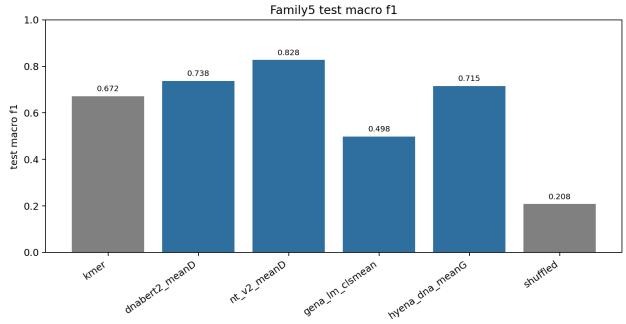

Fig. A1. Best 5-way family classification macro-F1 per encoder, with CDS 4-mer and shuffled-label baselines.

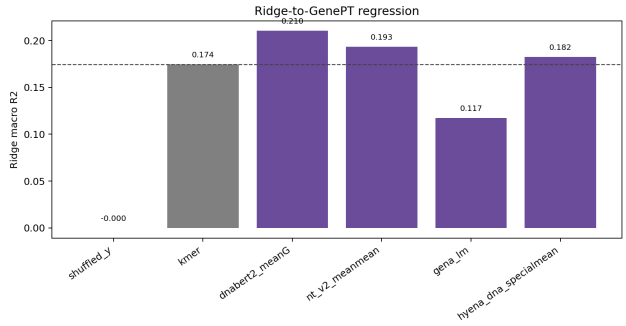

Fig. A2. Best Ridge-to-GenePT macro-R^2 per encoder, with the CDS 4-mer baseline.

A.2 Pooling and Regression Full Matrices

Table A3. All cached encoder-pooling cells for 5-way family classification.

Encoder	Pooling	Macro-F1	κ	$\Delta\kappa$	Accuracy
DNABERT-2	base	0.6490	0.6356	-0.0668	0.7721
DNABERT-2	meanmean	0.7220	0.7091	+0.0067	0.8172
DNABERT-2	specialmean	0.7205	0.7051	+0.0027	0.8152
DNABERT-2	meanD	0.7380	0.7226	+0.0202	0.8234
DNABERT-2	meanG	0.7275	0.7066	+0.0042	0.8131
DNABERT-2	maxmean	0.6144	0.5817	-0.1207	0.7331
DNABERT-2	clsmean	0.6547	0.6616	-0.0408	0.7906
NT-v2	base	0.8031	0.7984	+0.0960	0.8727
NT-v2	meanmean	0.7997	0.7982	+0.0958	0.8727
NT-v2	specialmean	0.7977	0.7985	+0.0961	0.8727
NT-v2	meanD	0.8275	0.8214	+0.1190	0.8871
NT-v2	meanG	0.8257	0.8179	+0.1155	0.8850
NT-v2	maxmean	0.5865	0.6099	-0.0925	0.7639
NT-v2	clsmean	0.5927	0.5689	-0.1335	0.7269
GENA-LM	base	0.4940	0.4568	-0.2456	0.6550
GENA-LM	meanmean	0.4940	0.4568	-0.2456	0.6550
GENA-LM	specialmean	0.4829	0.4589	-0.2435	0.6674
GENA-LM	meanD	0.4843	0.4668	-0.2356	0.6653
GENA-LM	meanG	0.4853	0.4899	-0.2125	0.6961
GENA-LM	maxmean	0.3502	0.2325	-0.4699	0.5175
GENA-LM	clsmean	0.4982	0.4776	-0.2248	0.6858
HyenaDNA	base	0.7103	0.6883	-0.0141	0.8049
HyenaDNA	meanmean	0.7103	0.6883	-0.0141	0.8049
HyenaDNA	specialmean	0.7095	0.6925	-0.0099	0.8070
HyenaDNA	meanD	0.6988	0.6830	-0.0194	0.8029
HyenaDNA	meanG	0.7149	0.6944	-0.0080	0.8090
HyenaDNA	maxmean	0.5614	0.5315	-0.1709	0.7105
HyenaDNA	clsmean	0.1396	0.0000	-0.7024	0.5359

Source: `analysis/tables/pooling_sweep_family5.csv`.

Table A4. All cached Ridge cells with delta versus the CDS 4-mer baseline.

Feature source	Pooling	R^2 macro	Δ vs 4-mer	Mean cosine	α
Shuffled labels	–	-0.0004	-0.1746	0.9130	1000
CDS 4-mer	baseline	0.1743	0.0000	0.9306	0.01
DNABERT-2	base	0.1812	+0.0070	0.9313	10
DNABERT-2	meanmean	0.2029	+0.0286	0.9333	10
DNABERT-2	specialmean	0.2029	+0.0287	0.9333	10
DNABERT-2	meanD	0.2100	+0.0357	0.9340	10
DNABERT-2	meanG	0.2104	+0.0361	0.9340	10
DNABERT-2	maxmean	0.1606	-0.0136	0.9293	100
DNABERT-2	clsmean	0.1911	+0.0168	0.9322	100
NT-v2	base	0.1926	+0.0183	0.9324	10
NT-v2	meanmean	0.1932	+0.0189	0.9324	10
NT-v2	specialmean	0.1931	+0.0188	0.9324	10
NT-v2	meanD	0.1882	+0.0139	0.9319	100
NT-v2	meanG	0.1902	+0.0159	0.9321	100
NT-v2	maxmean	0.1355	-0.0387	0.9271	100
NT-v2	clsmean	0.1172	-0.0571	0.9251	100
GENA-LM	base	0.1173	-0.0569	0.9251	100
GENA-LM	meanmean	0.1173	-0.0569	0.9251	100
GENA-LM	specialmean	0.1124	-0.0619	0.9246	100
GENA-LM	meanD	0.1084	-0.0658	0.9244	100
GENA-LM	meanG	0.1118	-0.0624	0.9245	1000
GENA-LM	maxmean	0.0400	-0.1342	0.9178	1000
GENA-LM	clsmean	0.1093	-0.0650	0.9242	100
HyenaDNA	base	0.1822	+0.0079	0.9313	1
HyenaDNA	meanmean	0.1822	+0.0079	0.9313	1
HyenaDNA	specialmean	0.1822	+0.0080	0.9313	1
HyenaDNA	meanD	0.1818	+0.0075	0.9314	1
HyenaDNA	meanG	0.1792	+0.0049	0.9310	10
HyenaDNA	maxmean	0.1329	-0.0414	0.9266	100
HyenaDNA	clsmean	-0.0015	-0.1758	0.9129	0.01
TSS 4-mer	baseline	0.0413	-0.1329	0.9173	0.01
TSS NT-v2	meanmean	0.1174	-0.0569	0.9252	0.1
TSS NT-v2	meanD	0.0545	-0.1197	0.9190	10
TSS NT-v2	meanG	0.0605	-0.1138	0.9197	10
Enformer trunk	global	0.1389	-0.0353	0.9271	100
Enformer trunk	centre	0.1425	-0.0318	0.9275	1000
Enformer tracks	centre	0.0135	-0.1607	0.9171	1000

Source: `analysis/tables/regression_full.csv`. TSS and Enformer deltas are shown relative to the CDS 4-mer baseline to match the generated artifact.

A.3 Context Ablation Table

Table A5. CDS, TSS, and supervised Enformer context comparison.

Feature source	F1	κ	Accuracy	Ridge R^2
CDS 4-mer	0.6722	0.7024	0.8172	0.1743
CDS NT-v2 meanD	0.8275	0.8214	0.8871	0.1882
TSS 4-mer	0.2452	0.2050	0.5873	0.0413
TSS NT-v2 meanmean	0.4468	0.3754	0.6407	0.1174
Enformer trunk	0.5450	0.4392	0.6448	0.1425

Source: `analysis/tables/context_ablation.csv`. The Enformer row uses global trunk for 5-way family classification and centre-bin trunk for Ridge.

A.4 Supporting Visual Diagnostics

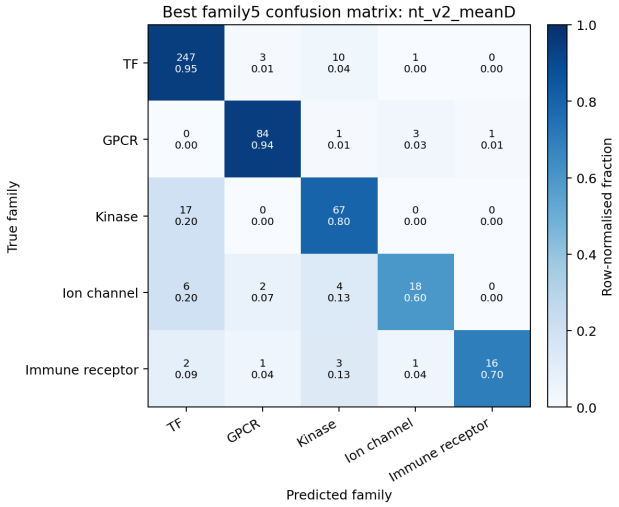

Fig. A3. Row-normalised confusion matrix for the best cached 5-way family classification encoder cell, NT-v2 `meanD`.

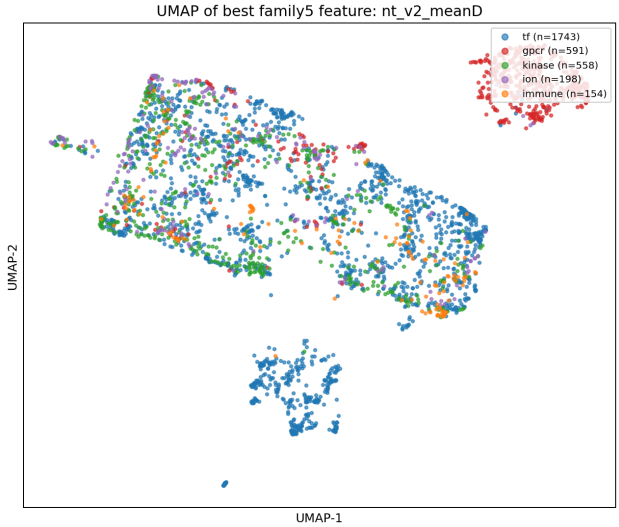

Fig. A4. UMAP (McInnes *et al.*, 2018) projection of the best cached 5-way family classification embedding table, NT-v2 `meanD`, coloured by family.

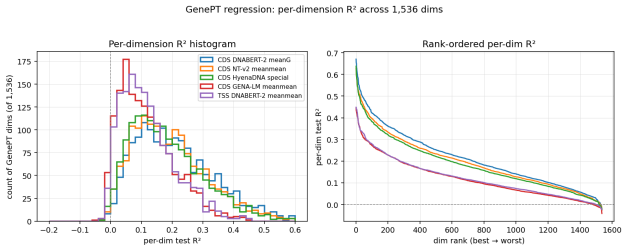

Fig. A5. Per-dimension test R^2 across the 1,536 GenePT output dimensions, for the four best CDS Ridge cells plus DNABERT-2 on TSS. Left: histogram of per-dim R^2. Right: rank-ordered per-dim R^2 (best dim on the left, worst on the right). For DNABERT-2 `meanG` CDS (macro-R^2 0.210), 1,527/1,536 dims have positive test R^2, the median per-dim R^2 is 0.190, and recovering 50% of the summed R^2 requires the top 429 dimensions (28% of the embedding space); NT-v2 `meanmean` CDS (macro-R^2 0.193) is essentially the same shape one band lower. The headline cross-modal R^2 is therefore distributed rather than concentrated in a handful of well-predicted summary axes. Source: `scripts/per_dim_r2.py` → `data/per_dim_r2.json`.